Vellore Institute of Technology
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BECE304L – Analog Communication Systems

Secure Li-Fi Based Door Locking System with IoT-Enabled Remote Access

Project Report

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ABSTRACT

Problem Statement

Traditional wireless door locking systems often rely on RF or Wi-Fi communication, which are susceptible to interception, interference, and security breaches. The growing demand for secure, energy-efficient, and low-cost smart access systems necessitates the exploration of alternative communication technologies.

Work Done

This project presents a novel Li-Fi-based password transmission system for door locking, eliminating the drawbacks of conventional RF-based solutions. A NodeMCU-based receiver receives an **ASCII-encoded** password transmitted optically via an LED on an Arduino-based transmitter. The system uses a synchronisation preamble and verifies the received password against a database of pre-authorised users. Additionally, a Blynk-based IoT interface enables remote password management and user access toggling. The system offers low latency, better physical security, and scalability for smart environments.

INTRODUCTION

Objective

The objective of this project is to develop a secure and efficient door locking system that utilises Light Fidelity (Li-Fi) communication to transmit password credentials. The system enhances security by transmitting encrypted data via light waves and integrates with an Internet of Things (IoT) platform, Blynk, to provide remote access and user management functionalities.

The proposed system addresses these concerns by leveraging Li-Fi technology—a form of optical wireless communication—for secure password transmission. Light-based transmission is inherently more secure, as it is confined to the physical line-of-sight and is less prone to external hacking attempts.

Relevance to SDGs

This project aligns with the United Nations Sustainable Development Goal (SDG) 9: “Industry, Innovation and Infrastructure” and SDG 11: “Sustainable Cities and Communities”. The secure and energy-efficient nature of Li-Fi supports sustainable technological infrastructure and enhances urban security systems.

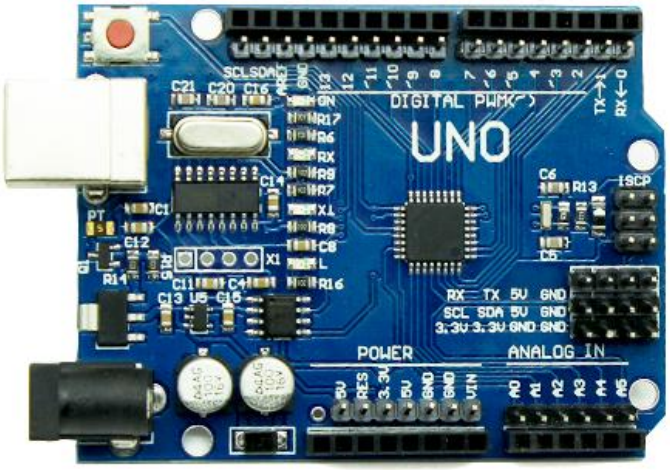
Key Contributions

- Designed a secure authentication system using Li-Fi communication with ASCII-based password encoding.
- Integrated Blynk IoT platform for remote password management and access control.
- Implemented a modular architecture using NodeMCU as the lock controller and Arduino as the password transmitter.
- Demonstrated low-cost, scalable implementation suitable for smart homes and secure facilities.

Component List

- **Transmitter Unit:** Arduino board, **LED** for transmission.

1. **Arduino UNO**
   A microcontroller development board based on the ATmega328P. It is used in the transmitter unit to control the LED and handle the encoding and timing of the data transmission. It features digital I/O pins, timers, and serial communication capability, making it ideal for simple control tasks.



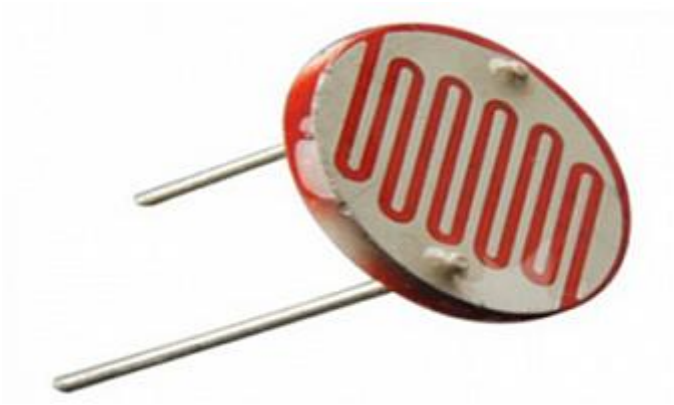
2. **Light Emitting Diode (LED)**
   An electronic component that emits light when current passes through it. In this project, it is used as a transmission medium in the Li-Fi system. The LED blinks at high speed to represent binary data corresponding to the password characters

- **Receiver Unit: NodeMCU (ESP32), LDR sensor, LED** for system status indication.

3. **NodeMCU (ESP32)**
   A Wi-Fi-enabled microcontroller board based on the ESP32 SoC. It is used in the receiver unit to handle password verification, system control, and internet connectivity. The board supports serial communication with the Arduino and integration with Blynk cloud services.

4. **Light Dependent Resistor (LDR)**
   A sensor whose resistance changes with the intensity of incident light. In this system, it is used to receive modulated light from the

transmitter LED. The changes in light intensity are interpreted as digital data by the microcontroller.



- **Additional Modules: WiFi module** integrated within NodeMCU and **blynk.cloud** for remote management.

5. **Wi-Fi Module (integrated in ESP32)**
   Provides wireless connectivity for the NodeMCU to communicate with the internet. It enables real-time data transfer to and from Blynk.cloud for remote user management.

6. **Blynk.cloud Platform**
   An IoT cloud platform that allows users to remotely interact with the NodeMCU. It supports features like password updates, access toggling for individual users, and emergency door unlock via a mobile app or web dashboard.

Methodology/System Overview

System Model

The proposed system integrates Light Fidelity (Li-Fi) communication with microcontroller-based access control and cloud-enabled remote management. It consists of three primary subsystems: (i) the optical transmitter (Li-Fi key), (ii) the receiver and verification module (door lock unit), and (iii) an Internet of Things (IoT) interface via Blynk Cloud for remote control and user management. The design is modular and scalable, allowing independent operation of each unit while maintaining seamless intercommunication.

The overall architecture ensures that sensitive authentication data is transmitted via optical medium, thereby mitigating the security vulnerabilities associated with conventional radio frequency (RF)-based systems. Passwords are encoded in ASCII and transmitted using visible light signals, which are then decoded and verified at the receiving end. A successful match enables physical access, while unsuccessful attempts are logged and rejected.

Proposed Methodology

The methodology is centred on secure short-range communication using Li-Fi for password transmission and NodeMCU for processing and validation. The three subsystems are elaborated below.

Transmitter Module (Li-Fi Key)

The transmitter module functions as a digital “key” that transmits the authentication password through modulated light signals. An Arduino board serves as the control unit and drives a visible spectrum LED to send binary data corresponding to each character in the password.

Initially, a preamble byte (0x2A, equivalent to binary 10101010) is transmitted to synchronise the receiver. This is followed by sequential transmission of ASCII-encoded characters from the pre-set password string. Each byte is transmitted with a fixed sampling interval, ensuring reliable decoding at the receiver. The LED remains in an idle state (ON or OFF) between transmissions to conserve energy and avoid false triggers.

Receiver Module (Li-Fi Lock)

The receiver unit is bifurcated into two distinct subsystems: (i) a Li-Fi signal decoder implemented using an Arduino board, and (ii) a NodeMCU microcontroller responsible for internet connectivity, access logic, and integration with the Blynk IoT platform.

The Arduino acts as the optical signal processor. It receives the modulated light signal via an LDR sensor, detects the preamble sequence for synchronisation, and sequentially reconstructs the transmitted ASCII characters corresponding to the password. Once the password is fully received, it is sent via UART (serial communication) to the NodeMCU for further processing.

The NodeMCU functions as the decision-making and control unit. Upon receiving the password from the Arduino, it verifies it against a list of stored credentials. These credentials can be remotely updated via the Blynk interface. The NodeMCU also handles:

- Enabling or disabling access for individual users.
- Changing the passwords associated with different users.
- Executing an emergency door unlock command if triggered remotely.

The modular division between sensing (Arduino) and control/networking (NodeMCU) enhances system reliability and simplifies debugging and upgrades. Moreover, this architecture separates real-time optical data acquisition from cloud-based logic,

ensuring smooth performance even during temporary network disruptions.

IoT Integration using Blynk

To enhance flexibility and user control, the system is integrated with the Blynk IoT platform. This allows real-time remote management of user access and password updates via a smartphone or web dashboard.

Key features include:

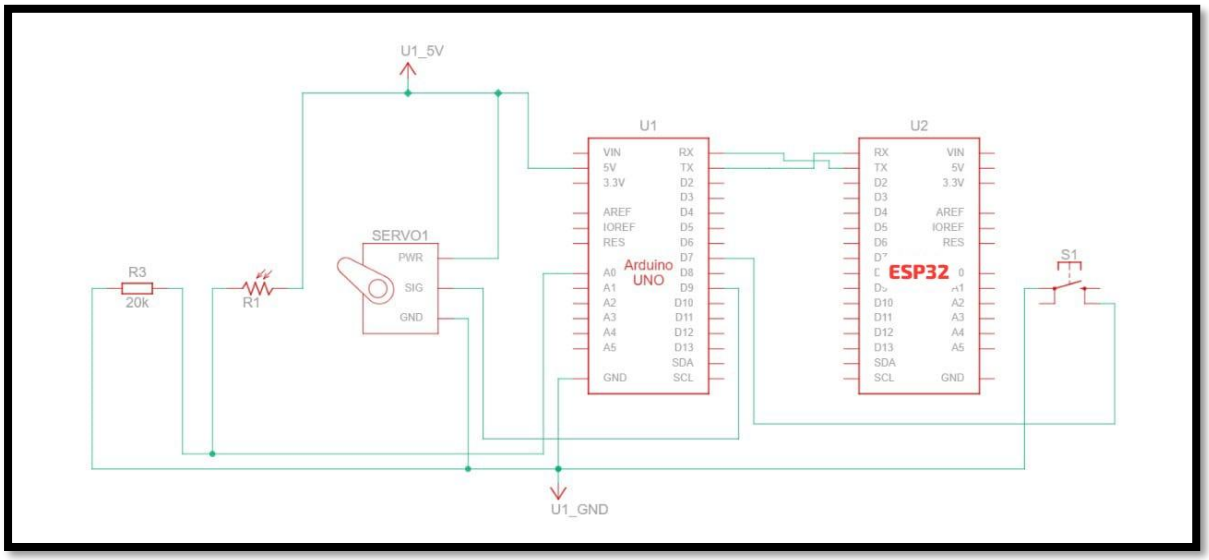
- Password Management: Each of the three user passwords can be modified remotely through virtual pins (V1–V3).
- Access Control: Administrators can enable or disable access for any user using toggle switches (V4–V6).
- Real-Time Feedback: System status, including authentication results and network connectivity, can be monitored via the Blynk interface.

This cloud-based functionality significantly enhances the adaptability of the system, enabling dynamic access control without requiring physical reprogramming of the device.

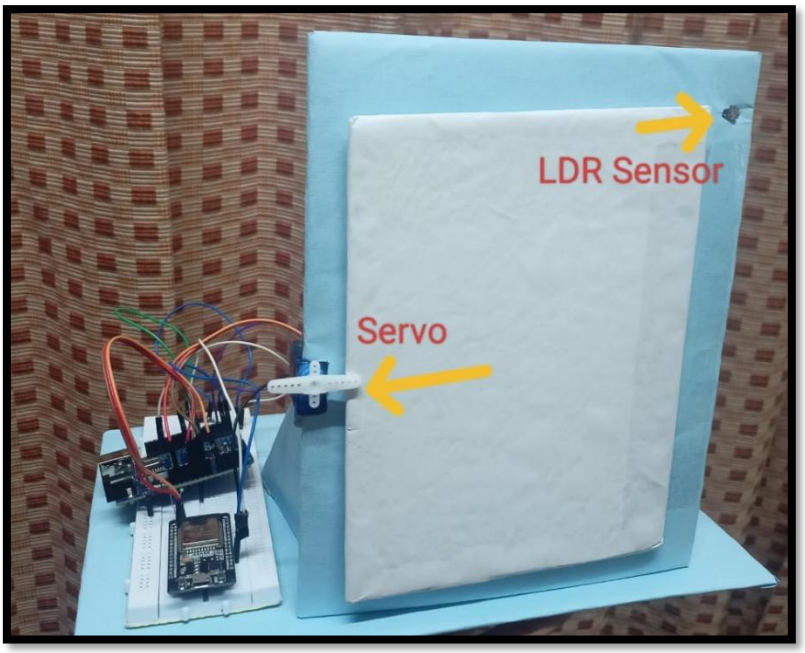
Security Considerations

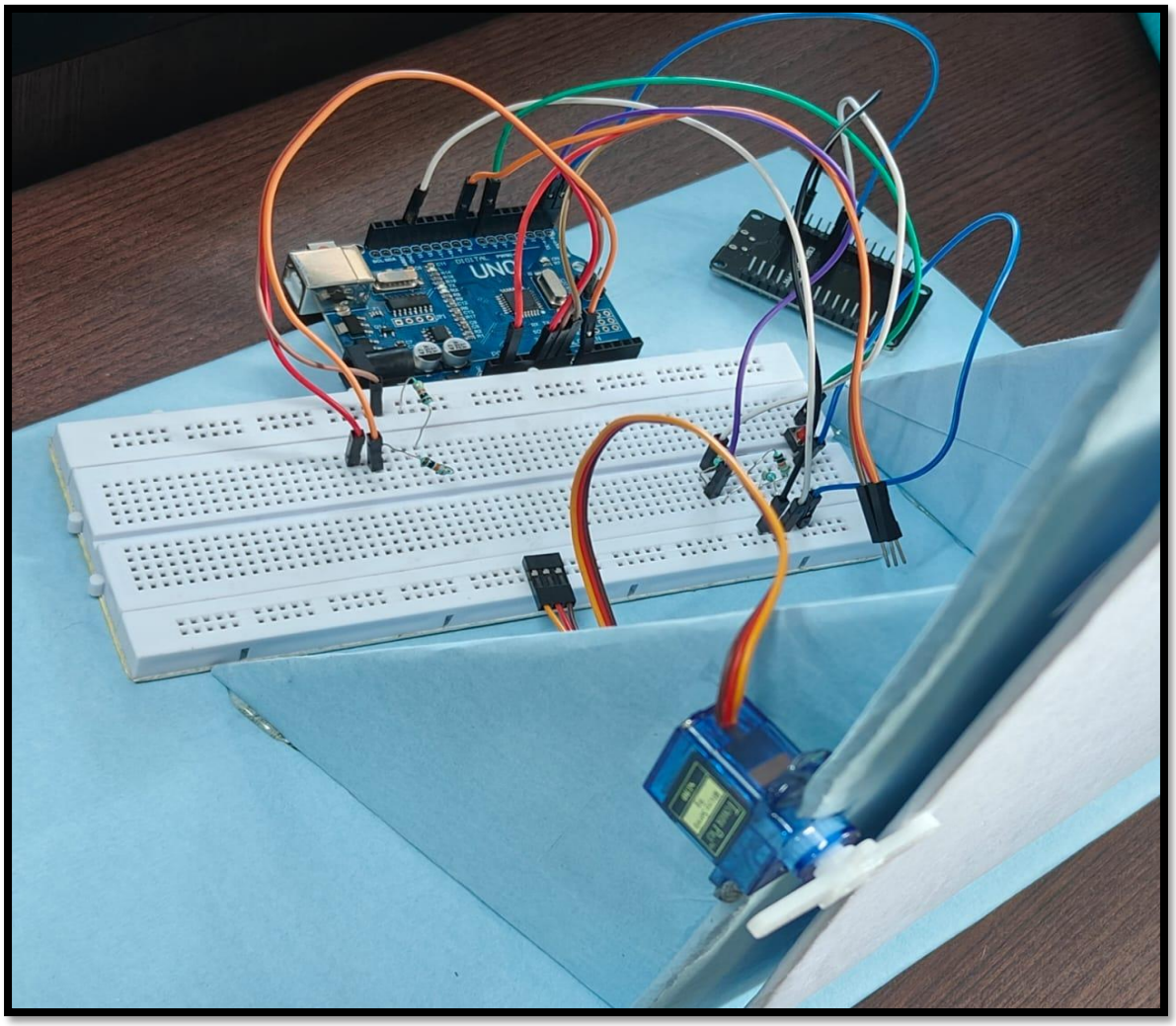
- Optical signal ensures data is confined within a room, unlike RF-based systems.
- Limited transmission range reduces chances of external hacking.
- IoT integration secured with user authentication tokens and password validation.

Circuit Diagram



Working Model





RESULTS & DISCUSSIONS

Results

- **Successful Li-Fi Link**: A stable Li-Fi communication channel was successfully established between the Arduino transmitter and the Arduino+NodeMCU receiver.
- **Accurate Data Decoding**: The receiver accurately detected preambles and decoded ASCII characters with a sampling interval of 40 ms.
- **Effective Password Validation**: Password verification was reliable within a 20–30 cm range under ambient lighting conditions.
- **Low-Latency Response**: The system responded to valid transmissions in under one second, ensuring real-time performance.
- **Remote Access and Control**: The Blynk app enabled seamless remote control, including password updates, user access toggling, and emergency overrides.
- **Stable Modular Communication**: By separating Li-Fi detection (Arduino) from Wi-Fi communication (NodeMCU), the system maintained stability even if one module encountered issues.

Discussion

- **Reliable ASCII Encoding**: Using ASCII character encoding streamlined the transmission process and improved decoding consistency.
- **Preamble-Based Synchronization**: Introducing a synchronization pattern (preamble) before each message significantly reduced transmission errors.
- **Ambient Light Mitigation**: External light sources caused minor interference with the LDR readings, but calibration effectively minimized this impact.
- **Division of Labor: Arduino + ESP32**:

    - The ESP32's multitasking nature and Wi-Fi stack introduced latency, making it unsuitable for timing-critical Li-Fi signal detection.
    - Splitting the system: Arduino was dedicated to real-time analog signal processing and door lock control, while NodeMCU handled IoT connectivity and cloud-based functions via Blynk.
- **System Reliability and Debugging**: This dual-controller setup simplified debugging and improved overall reliability by isolating responsibilities between hardware units.
- **Security and Access Control**: Password-based access control effectively prevented unauthorized entries, offering a secure door-locking mechanism.
- **Remote Management via Cloud**: Blynk's cloud interface supported remote user management and access configuration from anywhere in real-time.
- **Scalability and Future Integration**: The modular design supports future extensions such as:
    - Biometric authentication (e.g., fingerprint, facial recognition),
    - Encrypted communication for added security,
    - Li-Fi mesh networking for broader coverage,
    - Integration into smart home ecosystems.
- **Viable RF Alternative**: The system demonstrated its potential as a secure, low-interference alternative to traditional RF-based door locks, particularly in EMI-sensitive environments.

CONCLUSION

Project Summary

This project successfully demonstrates the feasibility and practicality of using Li-Fi (Light Fidelity) technology as a secure, low-cost alternative for short-range password transmission in smart door lock systems. By employing visible light through LEDs and encoding passwords in ASCII format, the system ensures robust performance and reduced susceptibility to electromagnetic interference compared to conventional RF-based systems.

IoT Integration and Flexibility

The integration of the Blynk IoT platform adds a significant layer of remote management capability. It allows real-time password updates, user access control, and emergency overrides, offering flexibility and ease of use in diverse environments such as homes, offices, and high-security facilities.

Enhanced Security and Reliability

The system's dual-controller design—dedicating Arduino to Li-Fi signal processing and NodeMCU to IoT communication—improves reliability by eliminating timing conflicts and isolating potential faults. Combined with password validation and future support for advanced authentication, the setup lays a strong foundation for secure smart infrastructure.

Contribution to Smart Infrastructure

By leveraging Li-Fi and IoT technologies, this system contributes toward building more secure, responsive, and sustainable smart environments. Its modular design ensures easy scalability and integration into existing smart ecosystems.

Future Work

To further enhance the system, the following developments are proposed:

- **Ambient Light Adaptation**: Implementing adaptive calibration algorithms to improve performance under varying lighting conditions.
- **Biometric Integration**: Incorporating biometric authentication methods (e.g., fingerprint or facial recognition) to strengthen access control.
- **Smart Home Expansion**: Extending functionality for seamless integration into larger smart home or building automation systems.

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